Transplantation and Cellular Therapy 28 (2022) 167.e1−167.e5

Transplantation and Cellular Therapy

journal homepage: www.tctjournal.org

ASTCT American Society for Transplantation and Cellular Therapy

Full Length Article
Infectious Disease

Safety and Efficacy of Prophylactic Levofloxacin in Pediatric and Adult Hematopoietic Stem Cell Transplantation Patients

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Article history:
Received 16 September 2021
Accepted 28 November 2021

Key Words:
Levofloxacin
Acute graft-versus-host disease
Clostridioides difficile
Hematopoietic stem cell transplantation
Multidrug-resistant organism
Bloodstream infection prophylaxis

A B S T R A C T

Levofloxacin has been widely used for bacteremia prophylaxis in the pre-engraftment setting for patients undergoing hematopoietic stem cell transplantation (HSCT), but data supporting this practice are inconsistent. In addition to concern for lack of benefit, there are also concerns that this practice could increase the rates of *Clostridioides difficile* (*C diff*) infections, the incidence of multidrug-resistant organisms (MDRO) or lead to increased incidence of acute graft-versus-host disease (aGVHD) by disrupting the gut microbiome. This study aimed to assess the safety and efficacy of levofloxacin as bacterial prophylaxis in pediatric and young adult patients undergoing allogeneic or autologous HSCT at a single pediatric center. We conducted a retrospective chart review evaluating patients age ≥6 months who underwent HSCT at our center between January 1, 2016, and July 31, 2020. Patients who underwent transplantation before March 2018 did not receive levofloxacin prophylaxis, whereas those who underwent transplantation after April 2018 did receive levofloxacin prophylaxis. Each transplantation was included as a separate episode if the patient underwent more than 1 transplantation during the inclusion time. The primary outcome of this study was the proportion of patients who experienced at least 1 bacterial bloodstream infection (BSI) in the first 100 days post-transplantation. Secondary outcomes included the number of non-levofloxacin antibiotic days post-transplantation, the incidence of aGVHD, the occurrence of *C diff* infections, and development of MDRO. A total of 370 HSCT recipients with a median age of 6.7 years (range, 0.5 to 39 years) were included in this study. Seventy-two patients had undergone more than 1 transplantation, and thus we had 443 transplantations to observe. Of these, 216 did not include levofloxacin prophylaxis and 227 included levofloxacin prophylaxis. There were no differences in baseline characteristics between the 2 groups except for age; patients in the non-levofloxacin prophylaxis group were younger (8.1 years vs 9.6 years; P = .05). There were no between-group differences in rates of death at 100 days, antibiotic use, fungal infections, or MDRO infections. Patients in the non-prophylaxis group developed more bacterial BSI in the first 100 days post-HSCT (27% versus 17%; P = .004) and more *C diff* infections (20% versus 9%; P = .003) than patients who received levofloxacin prophylaxis. In addition, more aGVHD was seen in the patients without levofloxacin prophylaxis (P = .014). Levofloxacin prophylaxis given from day -2 of HSCT through engraftment was significantly associated with decreased bacterial BSI in the first 100 days post-transplantation and was not associated with increased risks of *C diff*, aGVHD, or MDRO. Our study supports the use of levofloxacin prophylaxis in the peritransplantation period.

INTRODUCTION

Approximately 22,000 individuals undergo hematopoietic stem cell transplantation (HSCT) each year in the United States for a variety of underlying conditions, including malignancy, bone marrow failure disorders, primary immunodeficiencies, and metabolic disorders [1−3]. HSCT requires the administration of high-intensity chemotherapy-based conditioning regimens followed by the infusion of autologous or allogeneic progenitor cells. Despite advances in the field and supportive care, there is still significant treatment-related mortality and morbidity after transplantation, including the risk for opportunistic infections [3−5]. In fact, 20% of patients receiving an

Financial disclosure: See Acknowledgments on page 167.e4.
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https://doi.org/10.1016/j.jtct.2021.11.017

HSCT will sustain an infection post-transplantation, leading to death in some cases [4,6,7].

In the post-transplantation period, patients are routinely administered antimicrobial prophylaxis to prevent viral and fungal infections. The use of prophylactic transplantations in HSCT is not standardized or consistent. Patients undergoing HSCT are most susceptible to infections with gram-positive cocci; thus, fluoroquinolones, cephalosporins, and trimethoprim-sulfamethoxazole are common antibacterial agents used as prophylaxis [3,8]. At Cincinnati Children's Hospital Medical Center, the prophylactic use of levofloxacin in the periengraftment period has recently been adopted for all patients. Levofloxacin is a fluoroquinolone antibiotic that inhibits DNA-gyrase, or topoisomerase II, breaking DNA strands of bacterial organisms [9]. Levofloxacin covers gram-positive, gram-negative, and atypical organisms and is not marrow-suppressive. The American Academy of Pediatrics has identified several considerations of safety for the use of fluoroquinolones, including arthropathy, neuropathy, QTc prolongation, antimicrobial resistance, and *Clostridioides difficile (C diff)* infection [10].

Cincinnati Children's Hospital Division of Bone Marrow Transplant adopted prophylactic levofloxacin in all HSCT patients age ≥6 months beginning in 2018. Patients receive levofloxacin from day -2 until the absolute neutrophil count reached >200 cells/mm^3. This study aimed to determine the safety and efficacy of levofloxacin as bacteremia prophylaxis in HSCT recipients.

METHODS

We conducted a retrospective chart review evaluating patients age ≥6 months who underwent HSCT at Cincinnati Children's Hospital Medical Center between January 2016 and August 2020. Patients were excluded from this study if they were <6 months of age or had a documented allergy to a quinolone antibiotic (eg, levofloxacin, ciprofloxacin). Each transplantation was included and analyzed as a separate episode if eligible patients underwent >1 HSCT during the inclusion period. Patients in the levofloxacin prophylaxis group received levofloxacin according to our institutional standard, at a dose of 10 mg/kg twice daily (maximum 250 mg/dose) for patients age <5 years and 10 mg/kg once daily (maximum 750 mg) for patients age ≥5 years from day −2 until an absolute neutrophil count ≥200 cells/mm^3. If broad-spectrum antibiotics were started, then prophylactic levofloxacin was discontinued for the duration of empiric therapy and was reinitiated once empiric therapy was stopped.

This study was approved by the Institutional Review Board at Cincinnati Children's Hospital. The primary outcome of the study was the proportion of patients who experienced at least 1 bacterial bloodstream infection (BSI) in the first 100 days post-transplantation. Bacterial BSI was defined as the record of a positive blood culture with a subsequent course of antibiotic therapy. Patients with positive cultures who were not treated with a full course of antibiotics or with documentation that they were considered contaminants were not included in the study. Secondary outcomes included number of non-levofloxacin antibiotic days post-transplantation, incidence of aGVHD, proportion of patients experiencing at least one *C diff* infection, and development of multidrug-resistant organisms (MDRO). Antibiotic days were defined as the number of days the patient received antibiotics (excluding prophylaxis); if a patient received more than 1 antibiotic per day, more than 1 antibiotic day was counted. *C diff* infection was defined as a positive PCR followed by antibiotic therapy. Infections with MDRO were defined using the Center for Disease Control and Prevention's National Healthcare Safety Network antimicrobial-resistant phenotype definitions [11]. Demographic information, including primary diagnosis, transplant type, donor type, electrocardiography (ECG) results, aGVHD incidence, levofloxacin and antibiotic use, and *C diff* infections, were collected for this study.

Descriptive statistics were used to analyze the demographic data, which were then described with median and interquartile range. Fisher's exact test was used to compare categorical data between the two groups, and Wilcoxon's rank-sum test was used to compare continuous variables between the two groups. The incidence of bacterial BSI in the first 100 days post-HSCT for the first infection event was determined using the cumulative incidence function with death as a competing risk and compared using Gray's test. There were no relapses in the first 100 days in either cohort. Odds ratios with 95% confidence intervals were determined for outcomes between patients in the two cohorts. All statistical tests were 2-sided, and a *P* value <.05 was considered significant. Statistical analyses were conducted with SPSS version 28.0 (IBM, Armonk, NY).

Table 1
Baseline Demographics

Characteristic	Without Levofloxacin	With Levofloxacin	*P* Value
Age, yr, mean (range)	8.1 (0.5-38.7)	9.6 (0.5-39)	**.05**
Male sex, n (%)	117 (54)	121 (53)	.47
Race, n (%)			.44
Caucasian	186 (86)	201 (89)	
Black/African American	17 (8)	13 (6)	
Asian	4 (2)	8 (3)	
Black/Caucasian	9 (4)	5 (2)	
Diagnosis, n (%)			.18
Malignancy	100 (46)	127 (56)	
Bone marrow failure	35 (16)	25 (11)	
Immune deficiency	43 (20)	33 (15)	
Nonmalignant hematology	33 (15)	35 (15)	
Genetic/metabolic	5 (2)	7 (3)	
Graft source, n (%)			.16
Bone marrow	91 (42)	76 (33)	
Peripheral blood	113 (52)	141 (62)	
Cord blood	12 (6)	10 (5)	
Conditioning type, n (%)			.08
Myeloablative	159 (74)	184 (81)	
Reduced intensity	57 (26)	43 (19)	
Donor type, n (%)			.88
Autologous	54 (25)	60 (26)	
Related	51 (23)	53 (24)	
Unrelated	111 (52)	114 (50)	
Repeat HSCT, n (%)	27 (13)	32 (14)	.63

RESULTS

Study Population

Three hundred seventy participants with a median age of 6.7 years (range, 0.5 to 39 years) and a total of 443 HSCTs were included in the study. Among these 443 transplantations, 216 performed between January 2016 and March 2018 did not include levofloxacin prophylaxis, and 227 performed between April 2018 and July 31, 2020, included levofloxacin prophylaxis. Levofloxacin was administered for a median of 10 days in the prophylaxis group. Baseline demographics (Table 1) were similar in the 2 groups except for age; participants in the levofloxacin prophylaxis group were a little older (9.6 years versus 8.1 years; *P* = .05). There were no differences in the underlying diagnosis, preparative regimens, or stem cell source between the 2 groups.

Infectious Complications

The cumulative incidence of at least 1 bacterial BSI in the first 100 days post-HSCT was significantly higher in the no levofloxacin prophylaxis group (59 of 216 transplants) compared with the levofloxacin prophylaxis group (38 of 227; *P* = .004) (Figure 1). In the no prophylaxis group, in the first 100 days 157 patients (72.7%) had no BSIs, 43 (20%) had 1 BSI, 9 (4.2%) had 2 BSIs, and 7 (3.2%) had ≥3 BSIs, for a total of 97 bacterial BSIs. In the levofloxacin prophylaxis group, 189 patients (83.3%) had no BSIs, 24 (10.6%) had 1 BSI, 3 (1.3%) had 2 BSIs, and 11 (4.8%) had 3 BSIs in the first 100 days, for a total of 79 bacterial BSIs. In addition, the incidence of *C. diff* infections

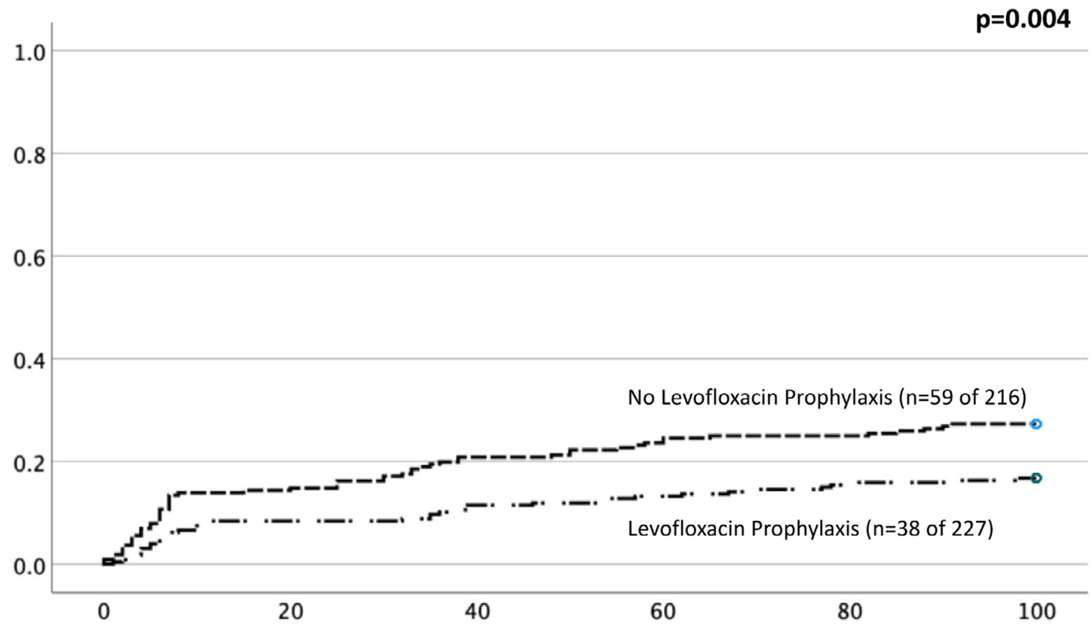

Figure 1. Cumulative incidence of at least 1 bacterial BSI in the first 100 days post-HSCT with death as a competing risk. Mortality in the first 100 days was similar in the patients who did not receive levofloxacin prophylaxis (n = 17; 8%) and those who did receive levofloxacin prophylaxis (n = 10; 4%) (*P* = .16).

Table 2
Summary of Primary and Secondary Outcomes

Outcome in First 10 d	No Levofloxacin Prophylaxis (N = 216)	Levofloxacin Prophylaxis (N = 227)	OR	(95% CI)	*P* Value
Proportion of bacterial BSI, n (%)	59 (27)	38 (17)	0.54	(0.34-0.85)	**.004**
Developed *C diff* infection, n (%)	42 (19)	21 (9)	0.39	(0.23-0.69)	**.003**
Developed 1+ fungal infection, n (%)	4 (2)	1 (0.4)	0.26	(0.03-2.12)	.21
Developed MDRO infection, n (%)	7 (3)	5 (2)	0.67	(0.21-2.15)	.50
Grade 2-4 aGVHD in allogeneic, n (%)	19 (12)	7 (4)	0.33	(0.14-0.80)	**.01**
Death within first 100 d post-HSCT, n (%)	17 (8)	10 (4)	0.96	(0.91-1.11)	.16
Death within first 12 mo post-HSCT, n (%)	51 (24)	41 (18)	0.80	(0.49-1.32)	.16
NRM in first 12 mo post-HSCT, n (%)	30 (14)	26 (11.5)	0.97	(0.91-1.04)	.48
Total antibiotic days, mean (range); *median*	47 (0-173); *38*	35 (0-177); *23*			<**.001**
Days of meropenem, mean (range); *median*	4.2 (0-44); *0*	2.5 (0-49); *0*			**.02**
Days of metronidazole, mean (range); *median*	1.4 (0-27); *0*	0.25 (0-15); *0*			<**.001**
Days of cefepime, mean (range); *median*	19.6 (0-78); *17*	14.9 (0-70); *12*			<**.001**

Significant *P* values are in bold type.
OR indicates odds ratio; CI, confidence interval; NRM, nonrelapse mortality.

was higher in the no prophylaxis group compared with the prophylaxis group (*P* = .003) (Table 2). Only 5 patients developed an invasive fungal yeast infection in the study, with no between-group difference in incidence (Table 2). There was also no difference in the incidence of MDRO infections between the 2 groups. Thirteen patients sustained an infection with an MDRO, including 7 patients (3%) in the no prophylaxis group and 5 (2.2%) in the levofloxacin prophylaxis group (*P* = .50); organisms included carbapenem-resistant Enterobacteriaceae, vancomycin-resistant *Enterococcus*, methicillin-resistant *Staphylococcus aureus*, extended-spectrum beta-lactamase- producing organisms, and carbapenem-nonsusceptible *Pseudomonas aeruginosa*.

Antibiotic Exposure

The average number of antibiotic days differed significantly in the 2 groups, with an average of 47 antibiotic days in the no levofloxacin prophylaxis group and 35 antibiotic days in the levofloxacin prophylaxis group (*P* < .001). The no prophylaxis group had significantly greater meropenem, metronidazole, and cefepime use (*P* < .05) (Table 2).

Transplantation Outcomes

Nineteen of 162 allogeneic HSCT recipients (11.7%) in the no prophylaxis group developed grade II-IV aGVHD by day +100, compared with 7 of 167 recipients (4.2%) in the levofloxacin prophylaxis group (*P* = .01). Mortality in the first 100 days was similar in the 2 groups (no levofloxacin, n = 17, 8%; levofloxacin prophylaxis, n = 10, 4%; *P* = .16).

No significant between-group difference in deaths within the first 12 months post-HSCT was seen (51 of 216 transplantations [23.6%] in the no prophylaxis group versus 41 of 227 [18.1%] in the prophylaxis group; *P* = .16). In addition, there was no statistically significant difference in non-relapserelated mortality in the first year post-transplantation between the 2 groups (P = .48) (Table 2).

Cardiac Outcomes

Baseline ECG results were available for 190 of 216 HSCTs (88%) in the no prophylaxis group and for 224 of 227 HSCTs (99%) in the prophylaxis group. Seventeen of 216 patients (7.9%) in the no prophylaxis group had a prolonged QTc interval at baseline, compared with 15 of 227 patients (6.6%) in the prophylaxis group (P = .46). Follow-up ECG was available for 93 patients (43%) in the no prophylaxis group and 106 patients (47%) in the prophylaxis group, revealing a prolonged QTc interval in 21 patients in the no prophylaxis group and 16 in the prophylaxis group (P = .20).

DISCUSSION

Levofloxacin has recently been used for bacteremia prophylaxis in the pre-engraftment setting for patients undergoing HSCT, but data supporting this practice are inconsistent [3]. In addition, recent guidelines advise against routine antibiotic prophylaxis in pediatric patients undergoing HSCT [8]. However, Ziegler et al. [12] studied levofloxacin use in adults receiving autologous HSCT and reported benefits in reducing central line infection rates and incidence of neutropenic fever [12]. In contrast, in 2018, Alexander et al. [13] published their data on using levofloxacin as neutropenic fever prophylaxis in the pediatric population showing a benefit in patients with acute leukemia receiving intensive chemotherapy but no statistically significant benefit in patients undergoing HSCT [13]. Although our study population was more closely comparable to the population studied by Alexander et al., our results are similar to those of Ziegler et al., revealing a statistically significant decrease in the number of bacterial BSIs in patients who received levofloxacin prophylaxis compared with those who did not.

Opponents of antimicrobial prophylaxis in this setting have argued that prolonged use of levofloxacin could lead to unwanted side effects. Specifically, it has been theorized that implementing standardized antimicrobial prophylaxis carries an increased risk for *C diff* and pseudomembranous colitis infections [9]. Regardless of antimicrobial prophylaxis, studies have shown that up to 20% of patients undergoing HSCT experience *C diff* infections within 30 days of transplantation [15]. *C diff* alone carries significant morbidity and mortality but also has been shown to increase the risk for aGVHD [15]. In addition, disruption of the gut microbiome from antimicrobial use also has been shown to increase the risk of aGVHD, as described by Gavriilaki et al. [14]. In our study, the incidences of both *C diff* infection and aGVHD were higher in the no prophylaxis group compared with the levofloxacin prophylaxis group, suggesting that antimicrobial prophylaxis with levofloxacin is safe and does not increase the risk for aGVHD or *C diff* infection.

We recognize that decreases in the rate of aGVHD or *C diff* infection could be related to the use of prophylactic levofloxacin along with the implementation of other institutional quality improvement initiatives during the study period. Specifically, the bone marrow transplantation service increased the use of abatacept for aGVHD prophylaxis; 49 HSCTs in the levofloxacin prophylaxis group included abatacept as GVHD prophylaxis, compared with HSCTs in the no prophylaxis group. The Infection Control Department also implemented changes that affected *C diff* rates throughout the hospital as a whole. These changes might have contributed to the lower incidences of aGVHD and *C. diff* infection seen in our levofloxacin prophylaxis group.

Concern for the emergence of MDRO infections is another argument for avoiding the use of prophylactic antimicrobials in this setting. Predic et al. [16] recently found a correlation between patients with carbapenem-resistant Enterobacteriaceae infections and recent fluoroquinolone use, suggesting that using prophylactic levofloxacin potentially could increase the risk for this patient population to incur a drug-resistant infection, making the infection more difficult and costly to treat. Therefore, we found it essential to analyze the sensitivity data to identify the organisms responsible for our episodes of bacteremia. However, we did not find a correlation between levofloxacin use and increased MDRO infection rate.

With a decreased bacterial BSI rate in the prophylaxis group, we expected to see decreased antibiotic use in the prophylaxis group. In agreement with this, patients in the levofloxacin prophylaxis group were exposed to an average of 13 fewer antibiotic days of therapy compared with their no prophylaxis counterparts. Patients were exposed to levofloxacin for a median of 10 days as prophylaxis. Even this short exposure led to fewer later bacterial BSIs and less total exposure to broader-spectrum antibiotics such as meropenem. Less antibiotic exposure likely would exhibit less selective pressure, producing fewer antibiotic-resistant organisms and ultimately decreasing morbidity and mortality in this patient population.

To our knowledge, this is the largest single-center pediatric study evaluating the effects of levofloxacin prophylaxis in allogeneic and autologous HSCT recipients. In contrast to the previously published data, our data show that levofloxacin appears to be effective at preventing bacteremia in pediatric and adult patients undergoing HSCT. In addition, prophylaxis with levofloxacin did not increase the incidence of *C diff* infection, aGVHD, or infections with MDRO in this patient population and therefore also appears to be safe. Limitations of our study include its retrospective nature and study design that included historical controls and thus did not account for other institutional quality improvement initiatives that could have contributed to patient outcomes. Moreover, including patients who underwent more than 1 HSCT could be considered a limitation. Larger, prospective multi-institutional studies should be conducted to confirm the promising results seen in this study.

DECLARATION OF COMPETING INTEREST

There are no conflicts of interest to disclose.

ACKNOWLEDGMENTS

Financial disclosure: There are no financial relationships relevant to this article to disclose.

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